Project Management Process

● Is a controlled process of initiating, planning, executing, and closing down a project.
● The project management process, involves four phases:
1. Initiating the project
2. Planning the project
3. Executing the project
4. Closing down the project

Project planning

● This phase involves creating a detailed roadmap for how the project will be executed, monitored, and controlled.
● The second phase of the project management process, which focuses on **defining** clear, discrete **activities** and the work needed to complete each activity within a single project.
1- Describing project scope, alternatives, and feasibility. aims to understand the content and complexity of the project.
It includes:
● What **problem** or opportunity does the project address?
● What are the quantifiable **results** to be achieved?
● What **needs** to be done?
● How will success be **measured**?

2- Dividing the project into manageable tasks. it is critical during the project planning process. It includes **dividing** the entire project into manageable tasks and then logically **order** them to ensure a smooth evolution between tasks.
● The process is called work breakdown structure (WBS).
● It helps to produce Gantt chart.
● Gantt chart: Is a graphical representation of a project that shows each task as a horizontal bar whose length is proportional to its time for completion.

What are the characteristics of a task
● Can be done by **one** person or a well-defined **group**.
● Has a single and identifiable **deliverable**.
● Has a known method or **technique**.
● Has well-accepted predecessor and successor steps.
● Is **measurable** so that percent completed can be determined.

3- Estimating resources and creating a resource plan. The goal of this activity is to estimate resource requirements for each project activity and use this information to create a project resource plan.
● Themostwidely used method is called COCOMO (Constructive Cost Model), which uses parameters that were derived from prior projects of differing complexity.
● COCOMO uses these different parameters to predict human resource requirements for basic, intermediate, and complex systems (size of S/W).
◦ Product Attributes: Required software reliability, Database size, Product complexity. ◦ Hardware Attributes:- Execution time constraints, Memory constraints.
◦ Personnel Attributes:- Analyst capability, Programmer capability.

Constructive Cost Model

● A semi-detached software project with a size of **50,000 lines** of code (50 KLOC).
● cost driver ratings/ parameters:
●-Required reliability: High (1.15)
●-Database size: Nominal (1.00)
●-Product complexity: High (1.17)
●-Programmer capability: High (0.86)
●-Use of software tools: Nominal (1.00)
●-Required development schedule: High (1.10)
● **Software Cost Estimation Tools:** Tools like SEER-SEM, SLIM, and Costar incorporate COCOMO principles for cost estimation.

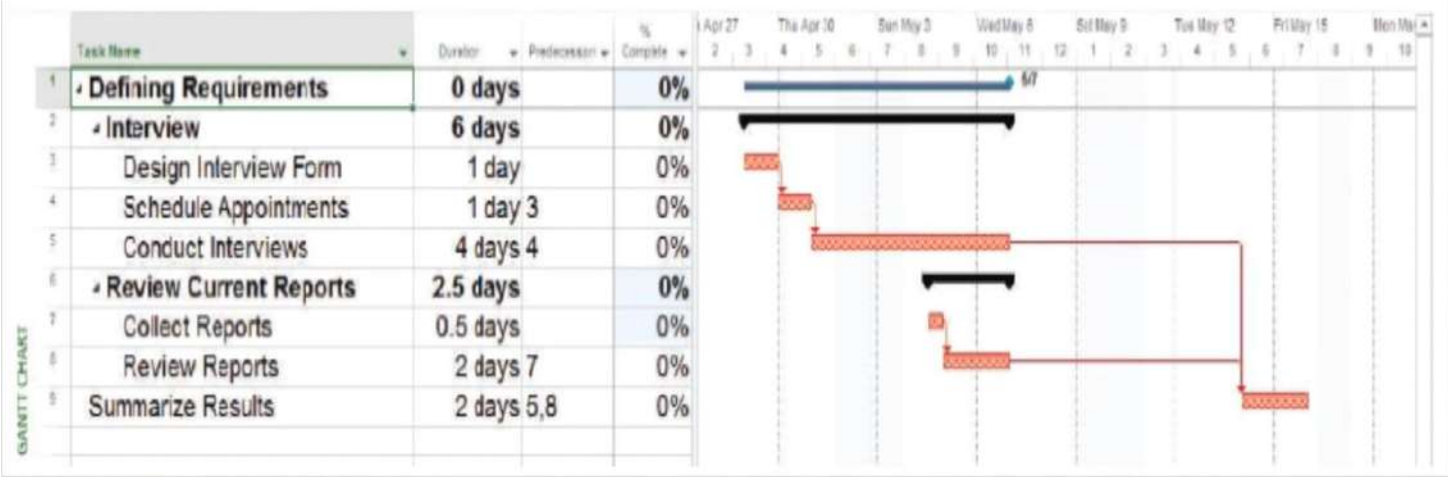
4- Developing a preliminary schedule. During this activity, you use the information on tasks and resource availability to assign time estimates to each activity in the work breakdown structure. These **time estimates** will allow you to create target starting and ending dates for the project.
● The schedule may be represented either as a Gantt chart or as network diagram.
● **Gantt Chart**: is a visual project management tool that helps teams plan, schedule, and track tasks over time. It displays tasks as **horizontal bars** along a timeline, showing their start and end dates, dependencies, and **progress.**
● Network diagram: Is a diagram that depicts project tasks and their **interrelationships**.

Gantt chart use case

Website Redesign Project: a company wants to redesign its website to improve user experience and modernize its branding. The project involves multiple tasks, including design, development, content creation, and testing.

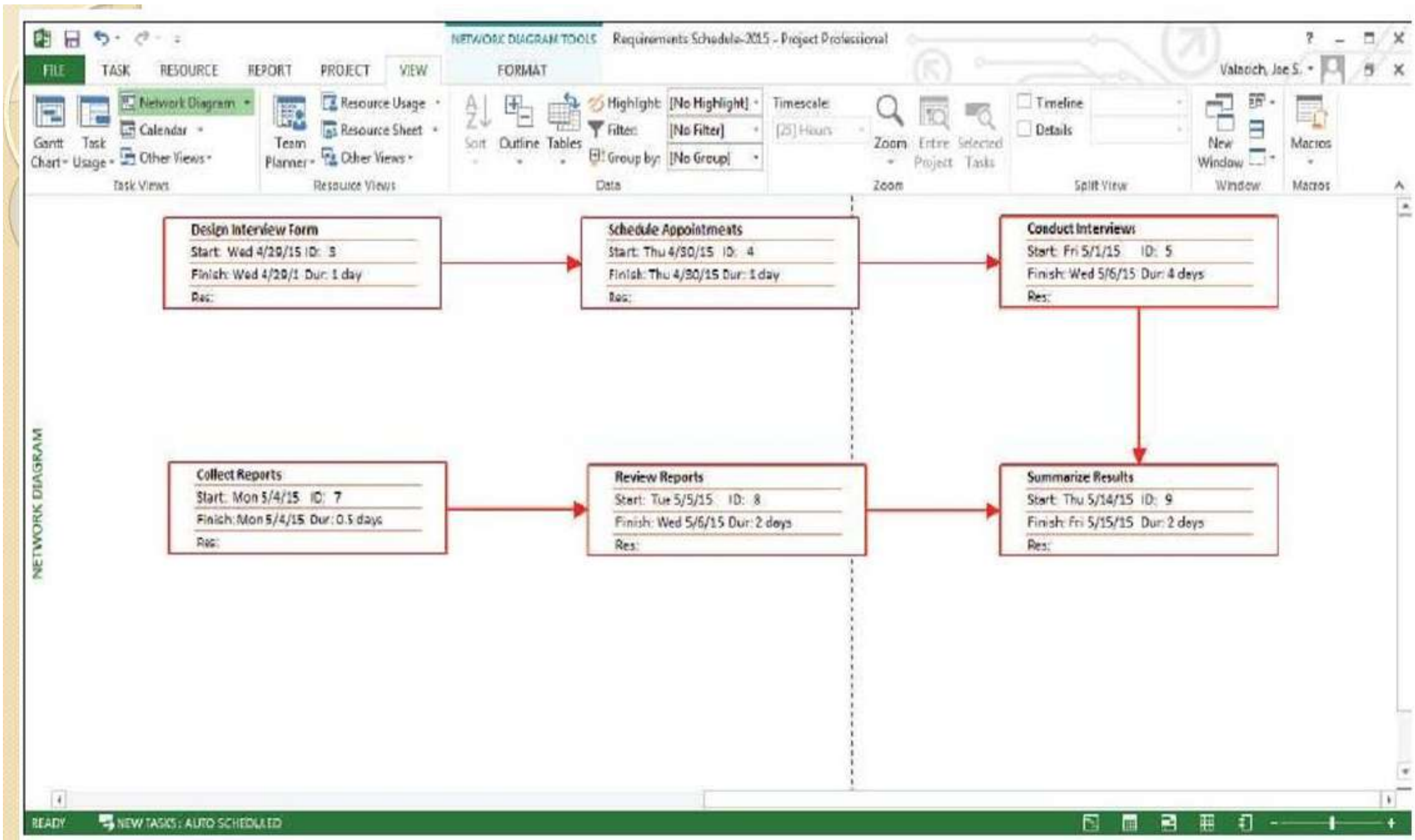
here is a breakdown of tasks and their timelines, represented in a Gantt chart:

Task	Start Date	End Date	Duration	Dependencies	Progress
Project Planning	Jan 1, 2024	Jan 5, 2024	5 days	None	100%
Gather Requirements	Jan 6, 2024	Jan 10, 2024	5 days	Project Planning	100%
Design Wireframes	Jan 11, 2024	Jan 20, 2024	10 days	Gather Requirements	100%
Design Mockups	Jan 21, 2024	Jan 31, 2024	10 days	Design Wireframes	100%
Frontend Development	Feb 1, 2024	Feb 15, 2024	15 days	Design Mockups	80%
Backend Development	Feb 1, 2024	Feb 20, 2024	20 days	Design Mockups	70%
Content Creation	Feb 5, 2024	Feb 25, 2024	20 days	Gather Requirements	60%
Integration and Testing	Feb 21, 2024	Mar 10, 2024	18 days	Frontend & Backend	50%
User Acceptance Testing	Mar 11, 2024	Mar 20, 2024	10 days	Integration and Testing	20%
Launch Website	Mar 21, 2024	Mar 31, 2024	10 days	User Acceptance Testing	0%



Gant Chart

● Tools for Creating Gantt Charts: Microsoft Project, Smartsheet (A cloud-based), Trello, Asana.
● Google Sheets/**Excel** (Manual creation of Gantt charts using spreadsheets)



Network Diagram

How the Gantt Chart is Used?
1. Task Scheduling: The Gantt chart clearly shows when each task **starts** and **ends**, helping the team stay on schedule.
2. Dependencies: It highlights task **dependencies** (e.g., Frontend Development cannot start until Design Mockups are complete).
3. Progress Tracking: The progress column shows how much of each task has been completed, allowing the project manager to identify **delays**.
4. Resource Allocation: The chart helps allocate resources efficiently by showing which **tasks overlap** and require simultaneous attention.

Project planning cont.

5- Developing a communication plan. The goal of this activity is to outline the communication procedures among management, project team members, and the customer. The communication plan includes:
1. when and how written and oral reports will be provided by the team.
2. how team members will coordinate work, what messages will be sent to announce the project to interested parties.
3. what kinds of information will be shared with vendors and external contractors involved with the project.

● Additionally, a project communication matrix that provides a summary of the overall communication plan can be developed. This matrix can be easily shared.
● Goal: to ensure that the **right people** are getting the right **information** at the right **time**, and in the right **format**.

Stakeholder	Document	Format	Team Contact	Date Due
Team Members	Project Status Report	Project Intranet	Juan Kim	First Monday of Month
Management Supervisor	Project Status Report	Hard Copy	Juan Kim	First Monday of Month
User	Project Status Report	Hard Copy	James Kim	First Monday of Month
Internal IT Staff	Project Status Report	E-mail	Jackie James	First Monday of Month
IT Manager	Project Status Report	Hard Copy	Juan Jeremy	First Monday of Month
Contract Programmers	Software Specifications	E-mail/Project Intranet	Jordan Kim	October 4, 2015
Training Subcontractor	Implementation and Training Plan	Hard Copy	Jordan James	January 10, 2016

Project Communication Matrix

● Determining project **standards** and procedures: during this activity, you specify how various deliverables are produced and tested by you and your project team.
◦ For example, the team must decide on which tools to use, how the standard SDLC might be modified, which SDLC methods will be used.
● Identifying and assessing risk: the goal of this activity is to identify sources of project **risk** and to estimate the **consequences** of those risks.
◦ Risks might arise from the use of **new technology**, prospective users' resistance.

● Creating a preliminary budget: during this phase, you need to create a budget that outlines the planned expenses and revenues.
◦ The project justification will demonstrate that the benefits are worth these costs.
◦ baseline plan will continue to be updated.
● Developing a project scope statement: an important activity that occurs near the end of the project planning phase is the development of the project scope statement.

Project planning cont.

- Setting a baseline project plan: once all of the prior project planning activities have been completed, you will be able to develop a baseline project plan.
- This baseline plan provides an estimate of the project's tasks and resource requirements and is used to guide the next project phase—execution.

Project Planning

1. Describing Project Scope, Alternatives, and Feasibility
2. Dividing the Project into Manageable Tasks
3. Estimating Resources and Creating a Resource Plan
4. Developing a Preliminary Schedule
5. Developing a Communication Plan
6. Determining Project Standards and Procedures
7. Identifying and Assessing Risk
8. Creating a Preliminary Budget
9. Developing a Project Scope Statement
10. Setting a Baseline Project Plan

Project Execution

- **Project execution** puts the baseline project plan into action.
- **Within the context** of the SDLC, project execution occurs primarily during the analysis, design, and implementation phases.

Project Execution

1. Executing the Baseline Project Plan
2. Monitoring Project Progress Against the Baseline Project Plan
3. Managing Changes to the Baseline Project Plan
4. Maintaining the Project Workbook
5. Communicating the Project Status

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Project Execution

- Executing the baseline project plan: as project manager, you oversee the execution of the baseline plan—that is, you initiate the execution of project activities, acquire and assign resources, orient and train new team members, keep the project on schedule, and ensure the quality of project deliverables.
- Monitoring project progress against the baseline project plan. While you execute the baseline project plan, you should monitor your progress.
  - If the project gets ahead of (or behind) schedule, you may have to **adjust resources, activities,** and budgets.
  - Monitoring project activities can result in **modifications** to **the current plan.**

- Managing changes to the baseline project plan: you will encounter pressure to make changes to the baseline plan.
- Communicating the project status. The project manager is responsible for keeping all team members—system developers, managers, and customers.
- **Two types of information exchanged** throughout the project:

(1) work results, or the outcomes of the various tasks and activities that are performed to complete the project.
(2) the project plan, which is the formal comprehensive document used to execute the project. The project plan contains numerous items including the project charter, project schedule, budgets, and risk plan.

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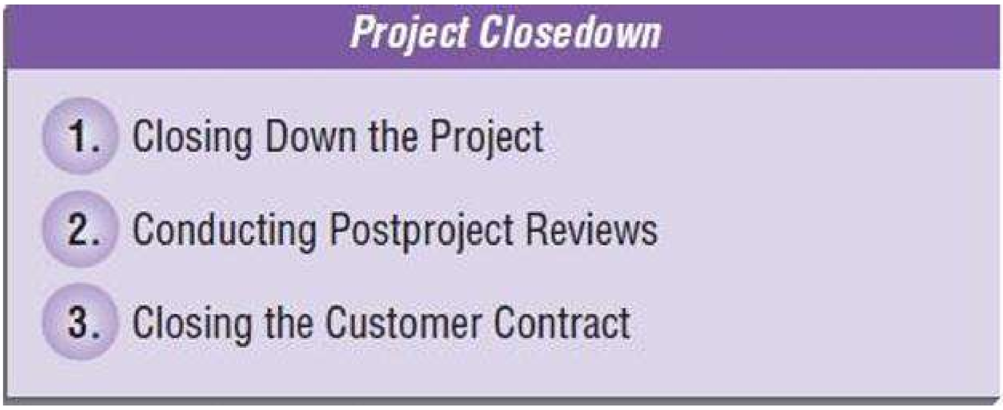
Closing Down the Project

● The final phase of the project management process, which focuses on bringing a project to an end.
● A **natural termination** occurs when the requirements of the project have been met—the project has been **completed** and is a **success.**
● An **unnatural termination** occurs when the project is **stopped before completion.**
● Within the context of the SDLC, project closedown occurs **after the implementation phase.**



● Closng Down. You will likely be required to assess each team member and provide an appraisal for personnel files and salary determination.

● Conducting post-project reviews. Once you have closed down the project, final reviews of the project should be conducted with management and customers. The objective of these reviews is to determine the strengths and weaknesses of project deliverables.

● Closing the customer contract. ensure that all contractual terms of the project have been met.